

Do speech foundation models perceive speaker similarity as humans do?

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1. Introduction

- Humans perceive speaker similarity on a continuous scale [1]
- Speech foundation models:
 - Extract speaker embeddings from speech
 - Use embedding similarity for speaker identification [2–4]
- Does speaker-embedding similarity align with human perception?
- If speaker embeddings align with human perception:
 - Human-like speaker perception may emerge from **data alone**
 - Such alignment may **guide perceptually grounded model design**
- Our objectives:
 - Analyze all Transformer layers of 43 models
 - Identify **model factors** associated with perceptual alignment

2. Method

- Graph representation of speaker similarity
 - Node set (speakers): S
 - Edge set (speaker pairs): $E = \{\{i, j\} \mid i, j \in S, i \neq j\}$
 - Edge weights: $W = \{c_{i,j} \mid i, j \in S, i \neq j\}$
 - $G = (S, E, W)$
- Perceptual speaker similarity
 - $c_{i,j}^{(\text{human})}$: Human-rated similarity score between the voices of speakers i and j
 - $G^{(\text{human})} = (S, E, W^{(\text{human})})$
- Speaker-embedding similarity
 - $c_{i,j}^{(\text{model})}$: Cosine similarity between the embeddings of speakers i and j .
 - $G^{(\text{model})} = (S, E, W^{(\text{model})})$
- Compare $G^{(\text{human})}$ and $G^{(\text{model})}$ using **three metrics**
 - Pearson correlation (LCC) $\uparrow$
 - Correlation between edge weights across unique speaker pairs
 - Frobenius distance $\downarrow$
 - Element-wise difference between the two similarity matrices
 - Spectral distance $\downarrow$
 - ℓ_2 distance between eigenvalues of normalized graph Laplacians
 - Captures differences in global graph structure

3. Experiments

3.1 Experimental setup

- Datasets
 - JVS (49 male, 51 female speakers) [5]
 - VCTK (52 female speakers) [6]
- Perceptual speaker similarity [5–7]
 - Human ratings from -3 to $+3$ for all same-gender speaker pairs
- Speaker-embedding similarity
 - For each layer, average all frames across 10 utterances per speaker to obtain a speaker embedding
 - Compute cosine similarity for all same-gender speaker pairs

→ Normalize both similarity scores to $[0, 1]$

3.2 LCC across all models and Transformer layers

- 43 open-source models [2–4, 8–17]
 - Supervised: ASR, TTS, TTA, audio classification; SSL: speech, general audio
- LCC varies across models: **high** for WavLM, **low** for Qwen3-TTS
- LCC **decreases with depth** in many models, but **increases** in some models (e.g., AudioGen)
- Perceptual alignment and its layer-wise trend may depend on **model configuration**

3.3 Model configuration: multiple regression

Note: **Bold** indicates coefficients with $p < 0.05$

Response variable	is_dec	is_mlang	is_ssl	hours	params	R^2
layer_max (maximum score across layers)						
LCC	-0.77	0.00	-0.14	-0.12	-0.14	0.747
Neg. Frobenius norm	-0.22	-0.02	-0.01	-0.01	0.05	0.983
Neg. spectral dist.	0.03	-0.16	-0.15	0.10	-0.03	0.835

- is_dec** < 0 : Encoder-based models show stronger perceptual alignment than decoder-based models
- is_ssl** < 0 : Supervised models show stronger alignment
- $R^2 \approx 0.8$ for layer_max, $R^2 \approx 0.2$ for layer_slope**: Peak alignment is **better explained** than layer trends

Response variable	is_dec	is_mlang	is_ssl	hours	params	R^2
layer_slope (slope across normalized layers)						
LCC	0.31	0.10	0.06	-0.26	0.29	0.205
Neg. Frobenius norm	-0.12	0.20	-0.02	0.06	0.07	0.123
Neg. spectral dist.	0.09	0.54	0.20	-0.15	0.13	0.241

- params**: Larger models have **lower peak LCC**, but **flatter layer-wise trends**
 - Decoder-based models show similar trends

3.4 Effect of fine-tuning on SSL models

- wav2vec2-{base,large}-960h
 - Compared with SSL-only models, scores **drop more sharply** with depth
 - ASR fine-tuning may **favor linguistic content**
 - **May suppress speaker-related information**
- wavlm-ssl_sv
 - Flatter** across layers than SSL-only WavLM
 - Fine-tuning for speaker verification may help **preserve speaker similarity structure**

3.5 Models trained on general audio

- audiogen-medium
 - LCC **increases monotonically** with depth
 - Audio generation training may **encourage deeper layers to encode finer acoustic details**
- atstframe_base
 - Relatively flat LCC** across layers
 - Without task specialization, the model may **retain acoustic representations across layers**

Reference

- [1] T. Kitamura, JASJ, 2015 [4] S. Chen, JSTSP, 2022 [7] Y. Saito, TASLP, 2021 [10] F. Kreuk, ICLR, 2023 [13] H. Hu, arXiv, 2026 [16] X. Li, TASLP, 2024
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